NATHISHWAR C

AI & ML Engineer | Gen AI | LLM Applications | RAG Systems | FastAPI | Python

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Portfolio | Linked in | Github

SUMMARY

AI & Machine Learning Engineering student with hands-on experience building production-ready AI applications, Retrieval-Augmented Generation (RAG) systems, LLM-powered solutions, and scalable backend services using Python and FastAPI. Experienced in developing adaptive learning platforms, semantic search systems, and intelligent recommendation engines with PostgreSQL, Redis, Docker, and FAISS. Passionate about solving real-world problems using Machine Learning, Generative AI, and scalable software architecture while continuously exploring modern AI technologies..

SKILLS SUMMARY

Programming: Python, SQL, JavaScript
Machine Learning: Scikit-learn, Feature Engineering, Model Evaluation , Data Preprocessing
Deep Learning: PyTorch, Neural Networks
Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, Hugging Face Transformers, Prompt Engineering
Backend: FastAPI, REST APIs
Databases: PostgreSQL, Redis, FAISS Vector DB
DevOps: Docker, Linux, CI/CD
Tools: Git, GitHub, Postman, VS Code

PROJECTS

KnowGrow AI — Adaptive Learning Intelligence Platform

- Designed and developed an AI-powered adaptive learning platform that personalizes learning using confidence-based spaced repetition algorithms.
- Engineered an intelligent recommendation engine that generates personalized study plans based on learner behavior and performance.
- Developed scalable REST APIs using FastAPI to manage authentication, course generation, quizzes, and learning progress.
- Integrated Large Language Models (LLMs) to generate learning content, quizzes, explanations, and AI tutoring assistance.
- Built Retrieval-Augmented Generation (RAG) pipelines to retrieve contextual educational content using vector embeddings.
- Utilized PostgreSQL, Redis, Docker, and FAISS to build a scalable backend architecture.
- Focused on modular architecture, maintainability, and API-first development principles.

AIKnowledge Assistant (RAG-based)

- Designed and implemented a Retrieval-Augmented Generation (RAG) system for intelligent document-based question answering.
- Built an automated document ingestion pipeline supporting embedding generation and semantic indexing.
- Implemented semantic search using FAISS Vector Database and Hugging Face embedding models.
- Developed scalable FastAPI backend services for chatbot interaction and document retrieval.
- Integrated LLMs with contextual retrieval to improve response relevance and reduce hallucinations.
- Optimized retrieval workflow for efficient document search across large knowledge bases.

WORK EXPERIENCE

AI/ML Intern — Google Developer Student Club (GDSC)

Oct 2024 – Dec 2024

- Developed Machine Learning models using Scikit-learn for predictive analytics and classification tasks.
- Built complete ML pipelines including data preprocessing, feature engineering, model training, evaluation, and inference.
- Worked with structured datasets to improve model performance through feature selection and preprocessing techniques.
- Collaborated with team members using Git-based workflows in an Agile development environment.
- Applied Machine Learning best practices for model development, experimentation, and evaluation.

EDUCATION

B.Tech Artificial Intelligence and DataScience — **9.0 CGPA**
Dhanalakshmi Srinivasan Engineering College — Aug 2023 - May 2027

Marutham matric.Hr.Sec.School — **89.8%**
Bio Mathematics — Jun 2021 - May 2023

PROJECT HIGHLIGHTS

- Built multiple AI-powered applications including RAG systems, adaptive learning platforms, and intelligent recommendation engines.
- Developed scalable backend services using FastAPI, PostgreSQL, Redis, and Docker.
- Designed Retrieval-Augmented Generation (RAG) pipelines with FAISS and Hugging Face embeddings.
- Built production-oriented REST APIs following modular software architecture principles.

CERTIFICATES

- AI Essentials - Google
- Machine Learning - Stanford & Deeplearning.ai
- Problem Solving — HackerRank
- Prompt Engineering — Google